Response to Reviewers for NATCHEMENG-24061044

Notes: Comments in blue | Replies in black | Actions in **bold**

Indicated page, line, figure, or reference numbers refer to the revised manuscript and/or supplemental information with changes highlighted.

Reviewer #1

There is not much to say about this manuscript: it is a very well written, comprehensive review about acetylene production with some comments also on downstream processes, including comments on sustainability and cost of the different pathways. As a review, it doesn't provide new discoveries, but it provides an interesting angle on a somewhat neglected topic. It is timely, since the chemical industry is facing a conversion process, in which acetylene could play a more important role again, and there are no recent comparable papers available, as far as I am aware. It has a very different focus compared to the review by Trotus et al., which is already ten years old. Within the scope of the review, all relevant aspects are adequately covered, the references are also appropriate. Thus, I suggest acceptance of the manuscript, I would require no changes. Whether this review format fits the scope of Nature Chemical Engineering as a recently launched journal is a question the editors have to answer - for me it would be an interesting supplement to the other papers that Nature Chemical Engineering carries.

Thank you for the highly positive feedback. We are delighted that the reviewer found our work comprehensive, timely, and offering a unique angle on the topic of acetylene production. We fully agree this is particularly pertinent now as the chemical industry is facing a conversion process towards more sustainable chemicals and revamping old chemical building blocks might bring a competitive edge. This was one of the main motivations of our study, as better highlighted in the revised introduction.

Reviewer #2

This review aims to provide a holistic overview of acetylene production technologies, proposes sustainable production initiatives and quantifies their economic and environmental performance, and explores potential applications. From my point of view, the starting point is that the authors seek to rekindle interest and foster collaborative efforts in advancing sustainable acetylene production and broadening its use as a feedstock for bulk chemicals. However, as a review paper submitted to Nature Chem Eng, the current form is still far from being considered for publication. Following are my specific comments on the review.

We would like to thank the reviewer for their feedback and helpful suggestions for improving our paper. Our aim is not only to summarize existing technologies, but to critically analyze existing challenges and, more importantly, present a data-supported strategy for green acetylene production. The contribution was widely appreciated as an innovative contribution by Reviewers #1 and #3. To better emphasize our focus, we have made the major revisions summarized below. We are convinced that the revised Perspective offers valuable insights for chemical engineers by envisioning a sustainable acetylene roadmap trajectory for C_2H_2 production and use, while encouraging innovation in its green production methods and downstream applications.

Overall actions: Prompted by the reviewer's comments to sharpen the main novelty of our contribution, we have transformed the manuscript from a 'Review' to a 'Perspective' with a new title, 'Towards green acetylene synthesis and application'. We have also modified the Introduction, significantly condensed the discussion on conventional C_2H_2 production technologies in Section 2, and added new emerging C_2H_2 application technologies in Section 3 to better reflect our forward-looking focus. Additionally, we have incorporated data on the predicted availability of renewable feedstocks, including biochar and biogas, by 2050, to further assess the feasibility of the proposed green C_2H_2 production roadmap. Detailed responses to the reviewer's comments are provided below.

2.1. As a review article, this manuscript spends the major part to discuss the conventional technologies to produce acetylene, such as CaC_2 route, partial oxidation and thermal plasma pyrolysis. To be noted, the acetylene production technologies nowadays all have no connections with catalysis. The reason would be attributed to the thermodynamic features of C_2H_2. The group of the authors are expertized in catalysis. I can't clearly see the C_2H_2-related research background of the authors, as well as the potential solutions to produce C_2H_2 via thermal catalysis route. Regarding the downstream organic synthesis of C_2H_2, I also cannot justify the new chemistry or new big market given from the review context. From my personal opinion, the authors for a review paper should have had deep understanding on such a research area with sound publications (i.e., research experience).

We disagree with the assertion that the authors expertise is limited to 'catalysis'. Our team also contains leaders in the areas of 'sustainability' and 'systems engineering' as evidenced by our publications. The corresponding author alone has published over 35 articles directly related to 'acetylene chemistry', in addition to extensive contributions on on 'alkane activation' and 'halogen chemistry', areas highly relevant to the chemistry discussed in this Perspective. Thus, we are confident in our qualifications to provide meaningful forward-looking perspectives in this field, a sentiment shared unreservedly by Reviewers #1 and #3. We encourage the reviewer to conduct a literature search before venturing into unsubstantiated comments.

We recognize that the original 'Review' structure, particularly the extensive discussion of conventional acetylene production technologies in Section 2, may have diluted the manuscript's central focus and overlapped with previous reviews. In response, we have restructured the manuscript as a Perspective and substantially streamlined Section 2 (see Action below for details) to better align with the future-oriented scope of the article. Technologically, while it is true that current commercial C_2H_2 production technologies are not catalysis-driven due to thermodynamic limitations, ongoing efforts in non-thermal

plasma and thermal pyrolysis approaches highlight the potential of catalysis in this domain. As discussed in the Outlook section, we propose that green C_2H_2 production could emerge in the near future from integrating green electricity and renewable carbon feedstocks, provided sufficient breakthroughs occur. Our Perspective does not claim catalytic thermal pyrolysis as the definitive path forward; rather, we seek to highlight the limited attention given to acetylene research by and inspire the catalysis community to explore this interesting field. We have also revised the manuscript to emphasize future directions for sustainable C_2H_2 production and its potential applications using emerging technologies such as electrocatalysis and photocatalysis, as well as novel C_2H_2 chemistries. Additionally, we underscore the increasing market demand for downstream chemicals like PVC (see Action below for details).

Action: We have significantly condensed Section 2 by merging Fig. 3 and 4 into a new Fig. 3, removing the previous Fig. 3a, 3b, 4b, and 4c, along with the associated discussions on conventional C_2H_2 production technologies. The remaining streamlined discussion in Section 2 provides a concise description for readers, helping them familiarize themselves with the field and understand our motivation on green C_2H_2 production and application. As suggested by Reviewer #3, we have incorporated recent advances in C_2H_2 applications into Section 3 (page 8, line 26–30). These include: (1) the conversion of C_2H_2 to butadiene (Nat. Catal. 2024, doi.org/10.1038/s41929-024-01250-0) or ethylene (C_2H_4) (Nat. Sustain. 2023, 6, 827; Nat. Catal. 2021, 4, 565) via electrocatalysis; (2) C_2H_2 hydrochlorination to VCM via photocatalysis (J. Am. Chem. Soc. 2024, 146, 29441); and (3) the coupling of C_2H_2 with CO into ester or acid products via thermocatalysis (Molecules 2024, 29, 1987; Angew. Chem. Int. Ed. 2023, 62, e202307570). Additionally, we have incorporated relevant discussions in Section 3 (page 8, line 21–23) regarding the expected 3.8% annual growth rate of PVC through 2031, that may accelerate the expansion of the acetylene market (data from Statista).

2.2. Some review papers on acetylene are well written in comparison with the current review manuscript, for example, Production of Acetylene and Acetylene-based Chemicals from Coal. Chem. Rev. 2014, 114, 3, 1743–1760; Catalytic Reactions of Acetylene: A Feedstock for the Chemical Industry Revisited. Chem. Rev. 2014, 114, 3, 1761–1782; Production of Acetylene from Viable Feedstock: Promising Recent Approaches. ChemPlusChem 2024, e202400247. https://doi.org/10.1002/cplu.202400247.

Following the superb feedback from Reviewers #1 and #3, we cannot understand on what basis the reviewer states that our study is less well written than the literature cited. While we agree that the referenced works are excellent and have contributed to the field, our manuscript offers a distinct perspective, as also highlighted by Reviewer #1, who explicitly recognizes its value in addressing a critical gap in recent literature.

Specifically, the review by Trotus et al. (Chem. Rev. 2014, *114*, 1761) primarily discusses catalytic reactions of acetylene but does not engage with modern sustainability challenges or explore pathways for green acetylene production. The review by Schobert (Chem. Rev. 2014, 114, 1743) focuses on coal-based C_2H_2 production, which is outside the primary scope of our work. Finally, the recent review by Gyrdymova et al. (ChemPlusChem 2024, e202400247), offers valuable historical insights but does not emphasize future technologies of green synthesis pathways, which are central to our manuscript.

Action: We have revised our work from a 'Review' to a 'Perspective' and updated the title to 'Towards green acetylene synthesis and application'. Additionally, we have significantly shortened Section 2 to reduce redundancy with the ChemPlusChem review mentioned by the reviewer. These revisions sharpen the focus of our work on the future green acetylene production, ensuring that it stands out distinctly from previous literatures.

2.3. The difference of this review from the existing reviews in literature would be the emphasis on the green footprint of acetylene production, e.g., changing coal to biochar. Considering the scale of acetylene production in market, this replacement (coal to biochar) is not convincing, and of course not realistic,

especially in China. A possible way to promote the green footprint would be the utilization of renewable energy generated electricity to drive the arc furnace or arc plasma for the high-temperature reaction environment in CaC_2 route or plasma pyrolysis of hydrocarbons.

We thank the reviewer for the insightful comment. While the conventional calcium carbide (CaC_2) route has high environmental impact, prompting many countries to transition from C_2H_2 to C_2H_4-based production, our Perspective aims to promote economically and environmentally sustainable acetylene production, particularly in markets outside China where these technologies may find greater adoption.

We acknowledge the reviewer's concern regarding the feasibility of using biochar and biomethane at scale. The current availability of biochar (0.4 Mt per year globally) and biomethane (0.03 Gt per year globally) is insufficient to meet the global acetylene demand. However, projections indicate that by 2050, the annual availability of biochar will range from approximately 0.3 to 0.5 Gt (Environ. Chem. Lett. 2021, 19, 3023). Considering that the CaC_2 process requires 2 kg of biochar per kg of C_2H_2 produced, rough estimates suggest that only 4 to 7% of the available biochar could meet the global C_2H_2 demand, estimated at about 11 Mt per year. Similarly, for biomethane-based C_2H_2 production pathways, specifically the partial combustion route, approximately 2.8 kg of biomethane is required to produce 1 kg of C_2H_2. It is projected that the biomethane potential within the European Union (EU) will reach around 0.1 Gt by 2050 (Biomethane production potentials in the EU, European Biogas Association 2022). Consequently, only 26% of this biomethane potential would be needed to meet the current C_2H_2 demand of 11 Mt. This number could be reduced even further with the adoption of plasma technologies. We acknowledge there will be competition for biochar and biogas with other sectors; however, the chemical industry is a hard-to-abate sector, making it a suitable candidate for preferential use of these feedstocks.

We also fully agree with the reviewer on the potential of using green or decarbonized electricity for enhancing the environmental sustainability of acetylene production. In this context, we highlight the potential future climate change impacts of C_2H_2 production under a decarbonized electricity grid in Fig. 6 of the main manuscript (purple scatter). It is important to note that these results reflect an optimistic scenario (i.e. achieving the 1.5°C target by 2050). We also recognize that these results are based on a global average, and do not capture the effects of variability in regional electricity grids.

Action: We have emphasized these aspects in Section 4 of the main manuscript (page 9, line 33–36; page 10, line 1–3).

2.4. The authors mentioned the major applications of C_2H_2 to synthesize VCM or 1, 4-butanediol. First, C_2H_2-VCM-PVC is indeed an important value chain, for instance, in China. However, this product chain can be totally substituted by C_2H_4-PVC route. One of the reasons for China to have to produce C_2H_2 for PVC is the balance of the chlorine from the chlor-alkali industry during the production of a large amount of alkali. Another reason would be the limited petroleum resources, which further limit the application of C_2H_4. Second, the technique to synthesize 1, 4-butanediol has been mature so that the production of 1, 4-butanediol is already serious overcapacity. Therefore, this part of the reviewed contents is not stimulating in the current and future world.

We agree that the C_2H_2-VCM-PVC route has largely been replaced by the C_2H_4-VCM-PVC route only outside of China, such as in Europe and the United States, where inexpensive petroleum ensures low-cost ethylene production. As the reviewer notes, in regions with limited petroleum resources, such as China, the acetylene route remains economically more viable, due to local constraints like the need to balance chlorine from the chlor-alkali industry. Importantly, the prominence of the C_2H_2-VCM-PVC route in China suggests a bidirectional shift between the C_2H_2- and C_2H_4-PVC production chains. Should green acetylene production become economically feasible and/or environmentally friendly, the C_2H_2-PVC route could regain relevance even outside of China. We believe that this warrants continued exploration of its potential in the context of green acetylene production technologies.

Concerning 1,4-Butanediol (BDO), we agree that global production is currently in a state of overcapacity, driven largely by the recent development of a more cost-effective C_2H_2-to-BDO process in China. This process has significantly reduced market prices and now accounts for over 55% of the world's capacity and 40% of global production (ChemCatChem 2020, 12, 2859; J. Adv. Manuf. Process. 2020, 2, e10054). While alternative routes using intermediates such as maleic anhydride or 1,4-diacetoxy-2-butene have gained prominence, the C_2H_2-based process remains critical in certain contexts, as demonstrated by its cost and efficiency advantages in China.

Action: We have included the reasons behind BDO overcapacity in Section 2.1 to highlight the significance of C_2H_2 (page 4, 30–33).

2.5. Regarding the environmental and economic assessment of acetylene synthesis routes, fossil based and green, I am not convinced by the replacement of methane by biomethane, coal by biochar as mentioned above. The production of C_2H_2 must consider where feedstock comes and how much volume of feedstock can be supplied sufficiently. Also, as mentioned above, C_2H_2 industry is closely connected with chlor-alkali industry. Therefore, the environmental and economic assessment of acetylene synthesis routes must be carried out in a specific chemical industrial park, or in a province/state, or across a country, even beyond. From my point of view, to improve the green footprint with acceptable environmental and economic assessment, the key is the green electricity, instead of the "green" feedstock, because of the process features.

As noted in our response to comment 2.3, while the current availability of biochar and biomethane is limited, projections indicate that by 2050, biochar (4% of its potential) and biomethane (26% or less of its potential) will be sufficient to meet current global C_2H_2 demands.

We recognize the reviewer's suggestion to evaluate acetylene synthesis in the context of specific industrial ecosystems, such as near the chlor-alkali industry. Our assessment assumes proximity between biochar, calcium carbide, and acetylene production facilities, thereby excluding transport-related impacts. However, as a representative case, we have calculated the climate change impacts including the transportation of biochar over a distance of up to 2000 km. Our results indicate that, regardless of the mode, transportation increases the climate change impacts of acetylene production by less than 5%. Similarly, for biomethane, we account for transport via pipelines, including emissions from potential system leakages, within our analysis. These findings affirm that transportation impacts are relatively minor, and our results remain robust across scenarios.

We agree with the reviewer on the pivotal role of renewable electricity in reducing the environmental footprint of C_2H_2 production. As highlighted in the response to comment 2.3, our analysis already includes a scenario based on decarbonized (green) electricity by 2050 (Fig. 6, purple scatter). Recognizing that no single solution fits all scenarios, we expand our focus beyond green electricity for sustainable C_2H_2 production to include renewable feedstocks. Furthermore, we stress that fully sustainable acetylene production would require shifting to renewable carbon as feedstock as the carbon content of C_2H_2 will be released after its use phase, contributing steadily to climate change. By using renewable carbon (e.g., from biochar or biomethane) as the main input, the carbon loop can be closed, removing the emitted carbon after the use phase and reintegrating it into the value chain. It is true, however, that closing the carbon loop is, in general, more expensive than using decarbonized power as input while still relying on fossil carbon as feedstock.

Action: The discussion in the main manuscript has been expanded to clarify the availability of biochar and biomethane (page 9, line 33–36; page 10, line 1–3). Additionally, Supplementary Note 1 of the Supplementary Information has been updated to include details on the system boundary for our economic and environmental assessment.

2.6. In the context, the major technical terms, such as partial combustion, hot plasma, thermal plasma, non-thermal plasma, cold plasma, warm plasma, thermal pyrolysis and so on, should not be confused.

Action: To ensure consistency and clarity, we have standardized the key technical terms into four categories: partial combustion, thermal plasma, non-thermal plasma (including cold and warm plasma), and thermal pyrolysis.

2.7. Figure 1 gives some historical information. However, why does not anticipate the near future, e.g., extend the timeline to 2050?

Alongside the introduction, Fig. 1 is designed to provide historical context, tracing the evolution of acetylene production and highlighting key milestones in its development. The purpose is to establish a foundation for discussing current challenges and opportunities for green C_2H_2 production. Adding speculative projections for 2050, while potentially interesting, would introduce significant uncertainties and detract from the focus on historical trends. The challenge in predicting future values arise from numerous uncertainties, including advancements in C_2H_2 production technologies, the development of competing renewable C_2H_4 production pathways, the availability of renewable carbon feedstocks, and differing environmental policies across countries.

Action: Instead of predicting the future of acetylene, we have incorporated a discussion on the anticipated availability of renewable carbon feedstocks for green C_2H_2 production by 2050, such as biochar and biomethane, in Section 4 (page 9, line 33–36; page 10, line 1–3).

2.8. Figure 2 provides the C_2H_2 production routes, involving past and current applications. However, trial routes and abandoned routes are not that justified. On feedstock, why not add the biomethane/biochar in? On products, why not predict possible market from the C_2H_2 chemistry for near future?

We appreciate the reviewers' questions. The inclusion of trial and abandoned production and application routes in Fig. 2 provides valuable historical context for understanding the evolution of C_2H_2 development, especially the challenges and barriers faced in developing acetylene technologies. As mentioned in our response to comment 2.7, predicting the future scale of C_2H_2 production and its market potential remains highly uncertain due to various influencing factors.

Action: Fig. 2 serves as a comprehensive analysis of both past and present C_2H_2 production and applications, with the associated challenges for each technology discussed in Section 2. These challenges highlight the need for exploring green C_2H_2 production in the future, such as using biomethane or biochar as feedstocks, which we address thoroughly in Fig. 4 and the corresponding discussions. We have also shortened the discussion of commercial and discontinued routes in Section 2.1. Moreover, we have included discussions on recent emerging C_2H_2 applications in Section 3 (page 8, line 26–30). Further details can be found in the response to comment 2.1.

2.9. Figure 3: in 3(a), some technical words are not appropriate, such as dryer, carbon, C_2H_2 furnace(?), $CO \rightarrow CO_2$(?), etc., 3(b) is not informative at all; in 3(c) the numbers are not representative, which should be specified with the corresponding C_2H_2 plant.

2.10. Figure 4: 4(a) has been in many existing references; 4(b) is a bit redundant with 4(a); 4(c) is not very informative at all.

In relation to the comments 2.9 and 2.10, the data in Fig. 3c are directly linked to an C_2H_2 plant in China (J. Clean. Prod. 2021, 322, 129055). We chose to retain Fig. 4a, despite its presence in other references, because the thermodynamic curve it presents is essential for helping readers, particularly outside the field, quickly grasp why C_2H_2 production from hydrocarbons at low temperatures is challenging.

Action: We have significantly shortened Sections 2.1–2.3 and consolidated Fig. 3c, 4a, and 4d into a new Fig. 3, removing all other components from the original Fig. 3 and 4. This adjustment reduces overlap with existing literature.

2.11. Figure 5: 5(a) is somewhat like a textbook or a slide of PPTs; 5(b) has little valued information; 5(c), such a plot (conversion vs. selectivity) can be often seen for comparing catalyst performance, however, it does not fit for C_2H_2 production via partial oxidation or thermal pyrolysis; 5(d) is not an appropriate evaluation for C_2H_2 production

We respectfully disagree with the reviewer's assessment of Fig. 4. Each panel addresses critical aspects of the technologies discussed:

- Fig. 4a showcases the reactor configurations of various non-thermal plasma technologies, emphasizing the complexity of reactor design and its pivotal role in advancing these systems.
- Fig. 4b illustrates a representative membrane reactor used in thermal pyrolysis to shift the methane (CH_4) conversion equilibrium towards the product side, demonstrating the importance of reactor design not only for non-thermal plasma but also for thermal pyrolysis in the future.
- Fig. 4c is intended to compare thermal plasma, non-thermal plasma, and thermal cracking processes, rather than commercial partial combustion technologies, which are addressed in Section 2.
- Fig. 4d uniquely presents CH_4 conversion, C_2H_2 selectivity, and energy consumption in a single plot, underscoring that energy requirements—often overlooked in non-thermal plasma studies—are as critical as conversion and selectivity for industrial viability.

We believe these figures collectively provide essential insights into the challenges and opportunities of emerging C_2H_2 production technologies and are crucial to the overall perspective.

2.12. Figure 6: most chemical engineers would not have interest in those "Potential green C_2H_2 synthesis routes."

In the context of the ambitious targets set by global frameworks such as the Paris Climate Agreement and the United Nations Sustainable Development Goals (SDGs), sustainability has become a central concern for all industries, especially the chemical industry. Given its heavy reliance on fossil fuels for both feedstock and energy, the chemical industry is one of the most difficult sectors to decarbonize, accounting for approximately 10% of global emissions (Curr. Opin. Chem. Eng. 2023, 39, 100881). As a result, the chemical industry is undergoing a major transition to achieve sustainable production by utilizing defossilized feedstocks and renewable energy. These potential green C_2H_2 synthesis routes align with the broader global imperatives mentioned above. We believe that this grand transition is both necessary and challenging, requiring collaboration across various disciplines, particularly with chemical engineers. Our experience, based on numerous interactions with the chemical industry and policymakers, indicates that the green chemicals transition is a top priority on the research agenda, with scientific literature on the topic growing exponentially.

Other comments:

2.13.1 Sections 2.2 and 2.3 discuss indirect methods of acetylene production (the calcium carbide method) and direct methods of acetylene production, are there any new developments of the methods to the research considered in the relevant reviews Schobert H. Production of acetylene and acetylene-based chemicals from coal[J]. Chemical Reviews, 2013, 114(3):1743-1760., Trotuş I T, Zimmermann T, Schüth F. Catalytic reactions of acetylene: a feedstock for the chemical industry revisited[J]. Chemical Reviews, 2014, 114: 1761-1782. a decade ago?

Action: We have consolidated Sections 2.2 and 2.3 into a single, concise Section 2.2, which provides a brief introduction to these technologies. The conventional commercial C_2H_2 production methods discussed in this section have experienced little to no significant development over the past decade. By

emphasizing the future rather than the past, this streamlined description serves as an essential primer for readers, helping them familiarize themselves with the field and understand our motivation for pursuing sustainable green C_2H_2 technologies. Additional details are addressed in the response to comment 2.1.

2.13.2. The title of section 2.3 (Direct commercial acetylene production routes) does not seem to be very appropriate, since the title of 2.2 (Indirect acetylene production route via calcium carbide) does not emphasize the word commercial. Therefore, this title may mislead readers unfamiliar with the relevant field into thinking that the calcium carbide method is not a commercial acetylene preparation method.

Action: We have significantly shortened Sections 2.1-2.3, and combined Sections 2.2 and 2.3 into a new section titled 'Commercial acetylene production routes.'

2.13.3. Emerging routes for acetylene synthesis are discussed in Section 2.4, where the authors describe a route for the production of acetylene from CH_4 using non-thermal plasma technology. Should this technique be viewed as a different way of generating plasma or as a revolutionary new technology for acetylene production? In the paper, the authors point out that the use of non-thermal plasma actually requires a greater energy input than the use of thermal plasma (Line 237), and that the CH_4 conversion and acetylene selectivity are not as good as in the thermal plasma approach. Thus, despite the savings in the quenching, non-thermal plasma technology should be far less economical than thermal plasma technology at this time. So, if this is viewed as a promising technology, what are the drivers for its development? What are the current key technical barriers? What technological breakthroughs are needed to industrialize it? At present the authors seem to be merely talk about the results reported in the literature, but not point out and analyze these issues clearly.

Non-thermal plasma is fundamentally different from thermal plasma in its generation process. Unlike thermal plasma, where all species (electrons, ions, and neutrals) are in thermal equilibrium, non-thermal plasma generates high-energy electrons while maintaining a relatively low bulk gas temperature. This characteristic potentially opens up new reaction pathways that are not accessible in thermal equilibrium systems. As shown in Fig. 4d, although non-thermal plasma requires a similar or even higher energy input compared to thermal plasma, its CH_4 conversion and C_2H_2 selectivity are already comparable to those of thermal plasma process. As an emerging technology that enables C_2H_2 production at low temperatures, we believe its potential should not be dismissed solely due to its slightly higher energy input compared to the current commercial thermal plasma process.

Action: We have elaborated our perspectives on non-thermal plasma in the Outlook section (page 12, line 8–15): Emerging technologies, such as non-thermal plasma, face commercialization challenges, requiring advancements in reactor design, catalyst development, and a deeper understanding of reaction mechanisms. It demonstrates performance close to commercial thermal plasma but typically require higher or similar energy input and more complex equipment. While catalysts can favor C_2H_2 production, their use does not appear to reduce energy requirements, and the design of catalytic plasma reactors remains complex and insufficiently explored. Therefore, future research should aim to improve C_2H_2 yield, energy efficiency, and equipment lifespan.

2.13.4. As far as I know, the production of acetylene from coal as a feedstock using plasma technology is also a promising technology and large-scale experimental installations have been established in China. However, only the acetylene production technologies based on natural gas as feedstock using plasma (thermal and non-thermal) are discussed in this paper. It is proposed to add a discussion in section 2.4 on the new developments and prospects of plasma technology for acetylene production from coal as a feedstock, in particular the new trends in recent years compared to what has been discussed in the previous reviews.

Action: To better highlight the emerging coal-to-C_2H_2 processes utilizing plasma technology, we have removed the relevant discussions from Section 2.1 and included the following description in Section 2.3 (page 6, line 5–10): Coal-to-C_2H_2 production via plasma technologies has seen significant development in China during the 21st century. Since 2014, Zhejiang University and Xinjiang Yuehetai Chemical Technology Co. Ltd. have collaborated to develop 2 MW and 10 MW plasma systems for C_2H_2 production (J. Theor. Appl. Mech. 2023, 55, 2779; Environ. Prot. Chem. Ind. 2021, 41, 229), claiming an annual production capacity of 5 kt, with costs over 20% lower than conventional CaC_2 routes and a significantly reduced carbon footprint.

2.13.5. In section 3, the authors argue that the adoption of bio-based feedstocks for the production of acetylene, and hence PVC, could be an important driver for the revival of acetylene. However, the fact that more than 45% of the market for PVC is held by China, and that this PVC is mainly produced from calcium carbide acetylene, implies that the ATV pathway would have had an importance comparable to that of EBP. Therefore, in terms of VCM or PVC production, it needs further consideration to whether the topic about the revival of acetylene exist. In addition, it is suggested that the production and environmental costs of the four methods in Figure 6 (i.e., fossil and green ATV or EBP) be put together in order to visualize the advantages of green ATV when combining production and environmental costs.

As the reviewer noted, China holds 45% of the global PVC market, where the ATV pathway dominates. The primary challenge associated with the ATV route is its high carbon footprint. Currently, nearly 20% of PVC in China is produced via the EBP route. If C_2H_2 production technologies shift from fossil-based to green alternatives, the EBP route could be partially replaced by the more cost-effective ATV route. More importantly, the ATV route may see a resurgence outside of China. The newly included discussions on emerging C_2H_2 applications further reinforce the potential for C_2H_2 to return as a key chemical building block.

Fig. 5a and 5b represent the production costs and climate change impacts of VCM production via both the fossil and green ATV or EBP, respectively. Presenting these data in two distinct figures enhances clarity and readability. Thus, in our view, combining them into a single figure would be suboptimal.

Action: We have incorporated discussions on recent emerging C_2H_2 applications into Section 3, as suggested by Reviewer #3 (see details in the response to the comment 2.1).

2.14.1. The second half of section 2.1 (Summary of acetylene production routes and its applications) puts too much emphasis on the substitution of acetylene applications, contrary to the main theme which calls attention to the reuse of acetylene. Suggests that the fact that acetylene applications have been replaced is coupled with the main advantages of using acetylene for synthesis and the problems that need to be overcome.

Our primary goal in Section 2.1 is to provide an extensive analysis of acetylene production and its applications, both past and present. By comparing C_2H_2's historical and current uses, we identify applications that have been replaced over time. This analysis also underscores the irreplaceable role of acetylene in certain applications, such as the production of VCM, butanediol, acetylene black, and vinyl acetylene, emphasizing its uniqueness. The primary challenge associated with acetylene-based routes remains their high carbon footprint, a concern discussed throughout the perspective.

Action: We have shortened Sections 2.1–2.3 and included the main advantages of using C_2H_2 as a building block in Section 2.1 (page 4, line 31–33). These advantages stem from its unique C≡C triple bond, which typically enables more streamlined downstream synthesis.

2.14.2. Section 2.4 (Emerging direct acetylene synthesis routes and catalytic applications) is rather informative and the refined title is not accurate enough. Acetylene from thermally cracked CH_4 is not part of the emerging direct acetylene synthesis route and the main focus of the presentation is on the

design of new catalysts for the TP process. It is suggested that plasma acetylene production and the development of new catalyst design for the TP process be presented in two separate sections.

Action: We have divided Section 2.4 into two subsections: (1) Emerging C_2H_2 synthesis routes via the plasma process, and (2) Advancements in catalyst design for the thermal pyrolysis process.

2.14.3. Section 5 (Environmental and economic assessment of acetylene synthesis routes) provides a more in-depth analysis of the environmental and economic evaluation of acetylene synthesis. However, processes related to acetylene inevitably need to consider safety as an equally important or even higher priority than economic and environmental aspects. It is proposed to supplement the evaluation of the safety performance related to the several synthesis methods mentioned with a summary of the risks and challenges faced by each process in terms of safety.

We fully agree that safety is a critical consideration in the synthesis of C_2H_2 (and generally in any other chemical process) due to its high flammability, explosive nature, and susceptibility to decomposition under specific conditions. Each C_2H_2 production process has unique safety challenges. For instance, in the calcium carbide process, CaC_2 must be stored in a dry, airtight container to prevent unintended contact with moisture. Similarly, in the partial combustion process, thermal runaway reactions can occur if the temperature is not carefully monitored and controlled. Finally, plasma-based processes for producing C_2H_2 require high-voltage electrical systems, which pose risks such as shock and electrical arcing. Therefore, appropriate safety measures must be implemented to ensure a safe and sustainable production process.

Action: A summary of the risks and challenges associated with each process is provided in Supplementary Note 3 of the Supplementary Information.

Reviewer #3

This review comprehensively introduced acetylene production technologies and provided extensive analyses regarding the possible ways to sustainably produce acetylene. Thus, it undoubtfully demonstrate the underlying and unexcavated value of acetylene. This review is expected to contribute to the clarification of the developing trend of acetylene in industry and advance relevant research with the basic guidelines and updated data. I recommend its publication in Nature Chemical Engineering after minor revisions.

We thank the reviewer for the strong endorsement. We appreciate their recognition of our extensive analysis towards sustainable acetylene production pathways, which effectively highlights its untapped potential. We are also pleased that this work is considered valuable in clarifying acetylene's evolving trends and advancing related research.

3.1. The authors planned a footprint of the revival of acetylene by only proposing the possibility of developing green acetylene, which seems not convincing enough. The authors may add extra part of discussion about the existing technologies or emerging technologies that may enlarge the application field of acetylene to make the plan more acceptable. For instance, recent work about the conversion of acetylene to ethylene using electrocatalysis way and to vinyl chloride using photocatalysis way provides the newest application field for acetylene.

Environmental and economic analyses in Section 3 demonstrate that vinyl chloride production could advantageously switch from the ethylene to the acetylene route if the biochar-based C_2H_2 process is industrialized. This paradigm shift would increase C_2H_2 utilization in VCM production, underscoring its expanding role in the chemical industry. As suggested by the reviewer, emphasizing emerging C_2H_2 applications using electrocatalysis and photocatalysis further highlights the potential of acetylene as a key building block for the future chemical industry.

Action: We have incorporated discussions in Section 3 (page 8, line 26–30) on recent studies that highlight emerging applications of acetylene: (1) the conversion of C_2H_2 to butadiene (Nat. Catal. 2024, doi.org/10.1038/s41929-024-01250-0) or C_2H_4 (Nat. Sustain. 2023, 6, 827; Nat. Catal. 2021, 4, 565) via electrocatalysis; (2) C_2H_2 hydrochlorination to VCM via photocatalysis (J. Am. Chem. Soc. 2024, 146, 29441); and (3) the coupling of C_2H_2 with CO into ester or acid products via thermocatalysis (Molecules 2024, 29, 1987; Angew. Chem. Int. Ed. 2023, 62, e202307570).

3.2. The authors briefly introduced several kinds of catalysts in part 2.4. What are the key issues for these catalysts during acetylene production, which hinders efficient acetylene production. It would be better to improve several perspectives about the current challenging problems in different catalysts and thus give some guidance for future research.

Various catalysts enhance C_2H_2 production in both non-thermal plasma and thermal pyrolysis processes; however, their efficiency is still insufficient to reach implementation at industrial scale. In the catalytic non-thermal plasma process, while C_2H_2 selectivity has already reached a high level comparable to that of commercial thermal plasma processes, the use of catalysts does not appear to reduce energy requirements. In fact, it often leads to energy consumption that is similar to or even higher than that of thermal plasma processes. Moreover, the design of catalytic plasma reactors for C_2H_2 production remains complex and insufficiently explored. On the other hand, in catalytic thermal pyrolysis, C_2H_2 selectivity is limited to ca. 35% due to the competitive formation of other hydrocarbons and coking. Consequently, catalytic thermal pyrolysis is not yet industrially viable, calling for research efforts on catalyst and reactor design.

Action: We have summarized the key challenges related to catalysis in these two technologies separately in Section 6 (page 12, line 8–25).